

Imperial Cambridge Math Contest 2019 Rd 1 — P2/6

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Find integers a and b such that

$$a^b = 3^0 \binom{2020}{0} - 3^1 \binom{2020}{2} + 3^2 \binom{2020}{4} - \cdots + 3^{1010} \binom{2020}{2020}.$$

Solution

Let $z = 1 + \sqrt{3}i = 2e^{i\pi/3}$. Then that sum on the right hand side is just $\Re(z^{2020}) = 2^{2020} \cos(2020\pi/3) = 2^{2020} \cos(2\pi/3) = (-2)^{2019}$.

Thus $a = -2$ and $b = 2019$ works. ■